NETWORKING PRESENTATION

OSPF

Open Shortest Path First

A Link-State Dynamic Routing Protocol

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1. What Is OSPF?

A simple definition

- OSPF stands for Open Shortest Path First.
- It is an Interior Gateway Protocol (IGP) used within an autonomous system.
- It allows routers to exchange network information and select efficient routes.
- It is an open-standard protocol, so it is supported by many networking vendors.

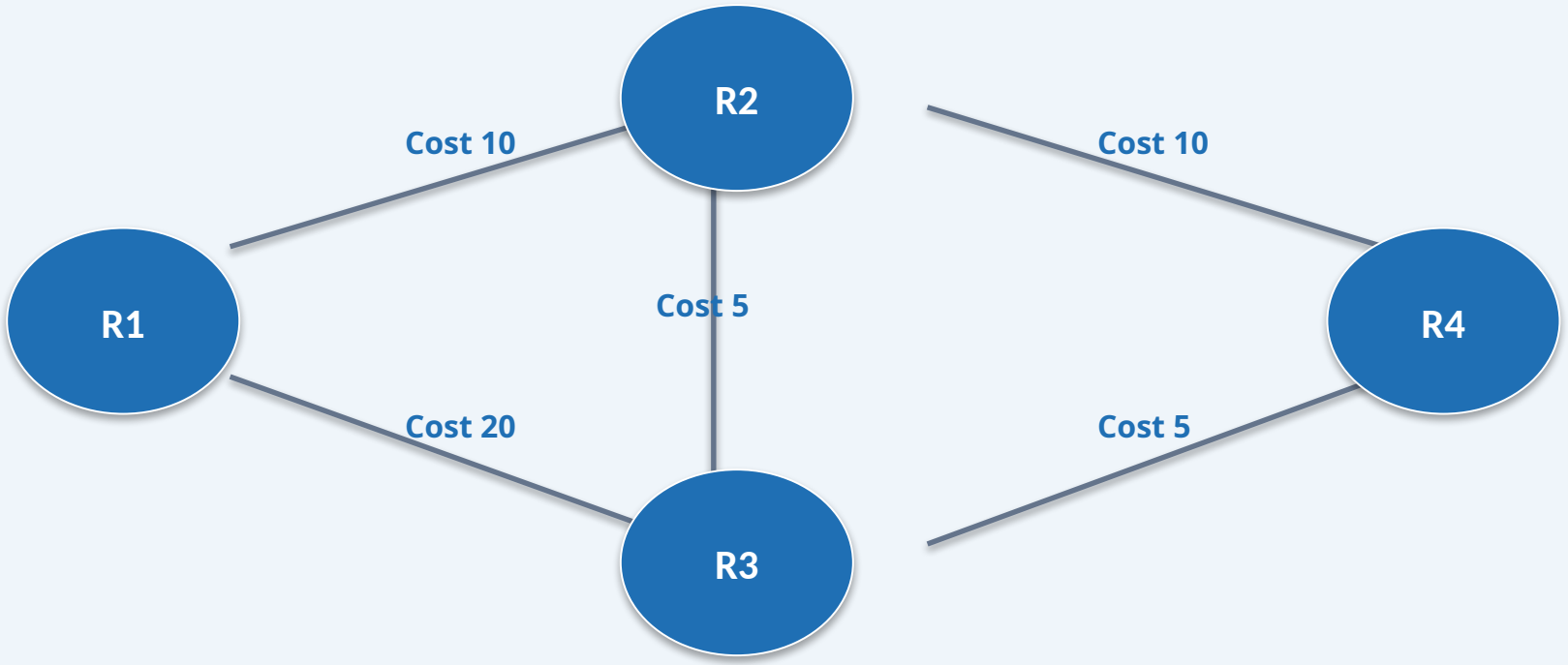
2. Why Do We Need OSPF?

The routing challenge

- A network may have several possible paths between two locations.
- Routers need a way to learn about available paths.
- OSPF calculates a preferred path using a metric called cost.
- If a link fails, OSPF can recalculate routes and adapt to the change.

3. How OSPF Works

A simplified routing process



1. Discover neighbors 2. Exchange link-state information 3. Build the topology database 4. Calculate the best paths

4. OSPF Route Calculation

Shortest Path First (SPF)

- OSPF uses Dijkstra’s Shortest Path First (SPF) algorithm.
- Each router builds a Link-State Database (LSDB) describing the network topology.
- The router calculates the lowest-cost path to each destination.
- The selected routes are added to the routing table.

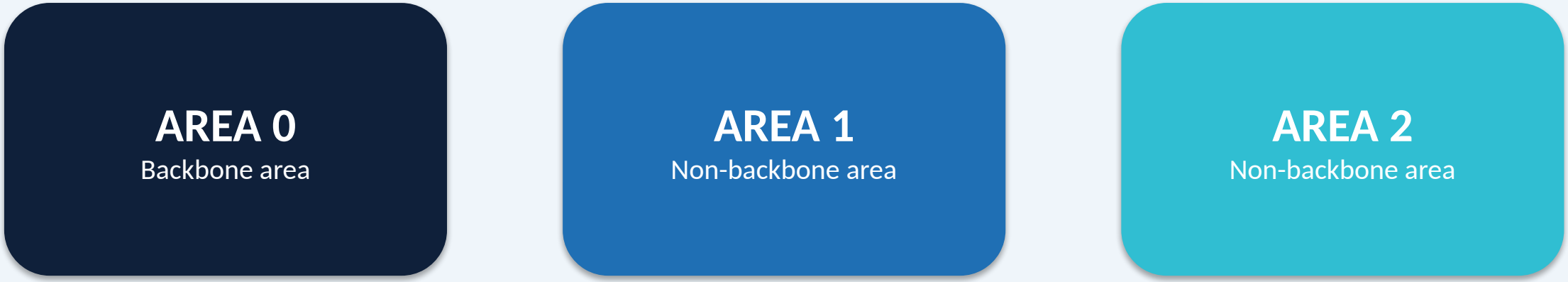
5. Important OSPF Terms

Key vocabulary

Neighbor	Another OSPF router discovered on a shared network.
Hello Packet	A packet used to discover and maintain neighbor relationships.
LSA	Link-State Advertisement; carries link-state information.
LSDB	Link-State Database; stores the collected topology information.
DR / BDR	Designated Router and Backup Designated Router on certain multi-access networks.

6. OSPF Areas

Organizing large networks



- Area 0 is the OSPF backbone.
- Areas help make large OSPF networks easier to manage and reduce routing information overhead.
- Area Border Routers (ABRs) connect areas.

7. Main Features of OSPF

Protocol characteristics

- Type: Link-State Interior Gateway Protocol.
- Metric: Cost, commonly based on interface bandwidth.
- Algorithm: Dijkstra’s SPF algorithm.
- Administrative Distance: 110 (Cisco default).
- IP Protocol Number: 89 (OSPF is not TCP or UDP).
- Supports hierarchical design through areas and supports route authentication.